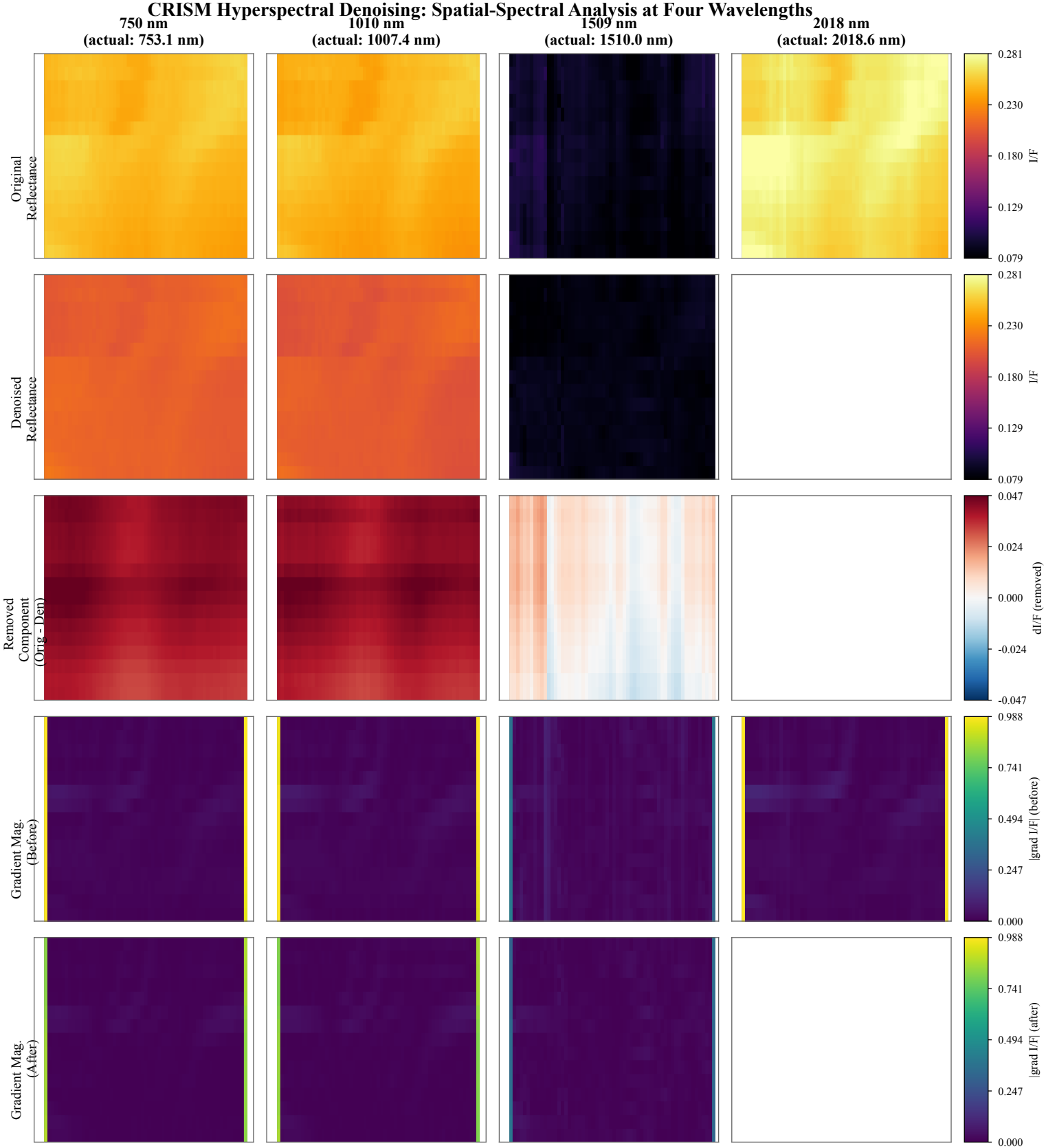


Data: MRO/CRISM TRR3 (FRT0001458F). Denoising: Stage-4 ML pipeline (Spectral DAE + Stripe CNN). Gradient: Sobel operator applied to each spatial band image.